

Module 07: Communication in Embodied

Learning Objectives

1. Define **Communication** within the context of Embodied.
2. Analyze how Communication interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of Communication.

Introduction

This module explores **Communication**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience.

Communication is a critical component of the 8-part Active Inference spine, bridging the gap between Learning and Planning.

Key Concepts

1. Communication as a Markov Blanket Boundary

How does Communication define the boundary between the agent and the environment?

2. Generative Models of Communication

What parameters involved in Communication must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Communication drive the perception-action loop?

Applications

In Embodied, we see Communication manifest in: * **Specific Example 1:** In partner dancing (tango, contact improvisation), the lead communicates the next movement not through words but through subtle shifts of weight, pressure through the hands, and directional intention felt through the torso; the follow's body receives these signals at the tactile Markov blanket and her generative model predicts the incoming movement trajectory, creating a shared predictive loop where two bodies communicate entire movement sequences through the embodied language of pressure, momentum, and spatial intention. * **Specific Example 2:** A basketball team executing a fast break communicates through coordinated movement -- a point guard's change of pace signals the wing to cut, a head fake communicates misdirection to the defense, and the timing of a bounce pass conveys expected arrival position; this movement-based communication operates through shared generative models where each player's body predicts the others' trajectories, creating a distributed active inference system where social coordination emerges from aligned embodied predictions rather than verbal instruction.

Conclusion

Understanding Communication allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.